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THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file sheaves.tex; no statement or proof step was rewritten, reordered or removed. The counted span is the author environment and its own proof environment. References inside the span are handed to the assembler as navigation stubs that carry the canonical tag of the cited tag only; whether each of them is load-bearing is recorded on the stub itself.

Allowed standard prerequisites (editorial).

- F#_x = F_x for every point (sheafification does not change stalks). invoked at statement line 1902 and proof lines 1926-1928. Identifies the left vertical components with stalks of F and supplies the injectivity of F_x -> Pi(F)_x.

Source reference. Original lemma 007Z: sources/stacks-sheaves/pinned-original-source.tex, source lines 1779–1779. Cited at statement line 1902 for the equality F#_x = F_x and at proof lines 1926-1928 for the injectivity of the bottom horizontal arrow; both uses load-bearing inside the counted span.

Source reference. Original section 007X: sources/stacks-sheaves/pinned-original-source.tex, source lines 1676–1676. Cited at statement lines 1904-1905 and 1915-1917 for the description of the horizontal maps; the description is load-bearing inside the counted span.

Lemma 0.2. *Let X be a topological space. Let $\mathcal{F}$ be a presheaf of sets on X. Let $U \subset X$ be open. There is a canonical fibre product diagram*

$$\begin{array}{ccc} \mathcal{F}^\#(U) & \longrightarrow & \Pi(\mathcal{F})(U) \\ \downarrow & & \downarrow \\ \prod_{x \in U} \mathcal{F}_x & \longrightarrow & \prod_{x \in U} \Pi(\mathcal{F})_x \end{array}$$

where the maps are the following:

1. *The left vertical map has components $\mathcal{F}^\#(U) \to \mathcal{F}^\#_x = \mathcal{F}_x$ where the equality is Lemma 007Z.*
2. *The top horizontal map comes from the map of presheaves $\mathcal{F} \to \Pi(\mathcal{F})$ described in Section 007X.*
3. *The right vertical map has obvious component maps $\Pi(\mathcal{F})(U) \to \Pi(\mathcal{F})_x$.*
4. *The bottom horizontal map has components $\mathcal{F}_x \to \Pi(\mathcal{F})_x$ which come from the map of presheaves $\mathcal{F} \to \Pi(\mathcal{F})$ described in Section 007X.*

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Proof. It is clear that the diagram commutes. We have to show it is a fibre product diagram. The bottom horizontal arrow is injective since all the maps $\mathcal{F}_x \to \Pi(\mathcal{F})_x$ are injective (see beginning proof of Lemma 007Z). A section $s \in \Pi(\mathcal{F})(U)$ is in $\mathcal{F}^\#$ if and only if $(*)$ holds. But $(*)$ says that around every point the section s comes from a section of $\mathcal{F}$. By definition of the stalk functors, this is equivalent to saying that the value of s in every stalk $\Pi(\mathcal{F})_x$ comes from an element of the stalk $\mathcal{F}_x$. Hence the lemma. $\square$